
SPE Rate Exclusion cut

April 2025 version

— Abantika & Ewan —
05/14/2025

Data Description:

[Previous version](#)

File used: WS runs (12205 - 19075) until April 17, 2025

Cuts applied:

- Muon veto
- E-train veto ← ***March collaboration meeting***
- GTV
- Spatial hotspot + HSE/S2
- PMT Trip

ALPACA branch: EDFspatialTest ← ***~April cut bundle***

Module used: [hotspotCheck](#) (Branch: EDF)

Update Strategy Summary:

- Adopted March 2025 E-train + GTV cuts
→ Removes ~16% of livetime
- Previously used WS2024 ETV
→ Removed ~10% of livetime
- Older ETV let more UDTs through
→ HSPE exclusions had to compensate downstream
- Updated veto reduces UDTs upstream
→ Enables less aggressive, more optimal downstream cuts

Rate Calculation:

$$\text{SPE rate (Hz)} = \sum \text{Number of SPE pulses} / \sum \text{Total event time in the bin [s]}$$

$$\text{S1 rate (Hz)} = \sum \text{Number of S1 pulses} / \sum \text{Total event time in the bin [s]}$$

$$\text{SPE_MPE rate (Hz)} = \sum \text{Number of SPE+MPE pulses} / \sum \text{Total event time in the bin [s]}$$

Optimization Strategy:

We do this using a 2D grid scan across two parameters:

- Threshold — how strict the cut is, expressed in sigma (σ) above the mean
- Binning — the time resolution over which rates are computed (e.g., 10s, 30s, 60s...)

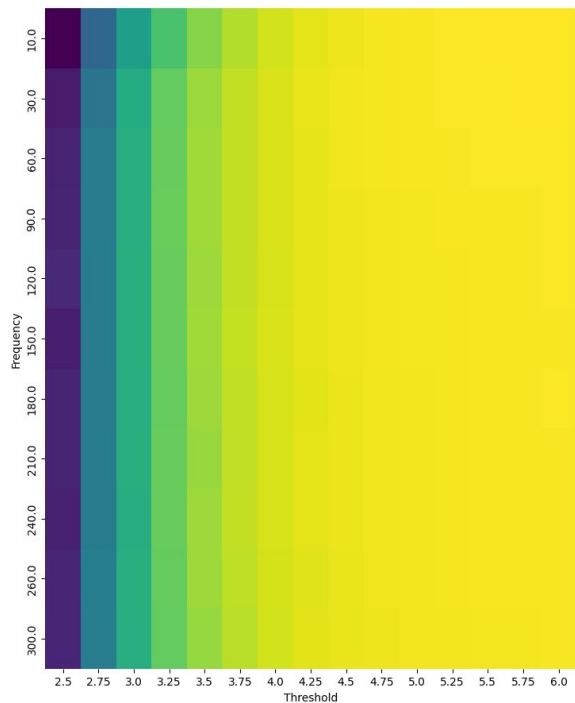
For each combination, we evaluate:

1. Livetime retained
2. Background retained

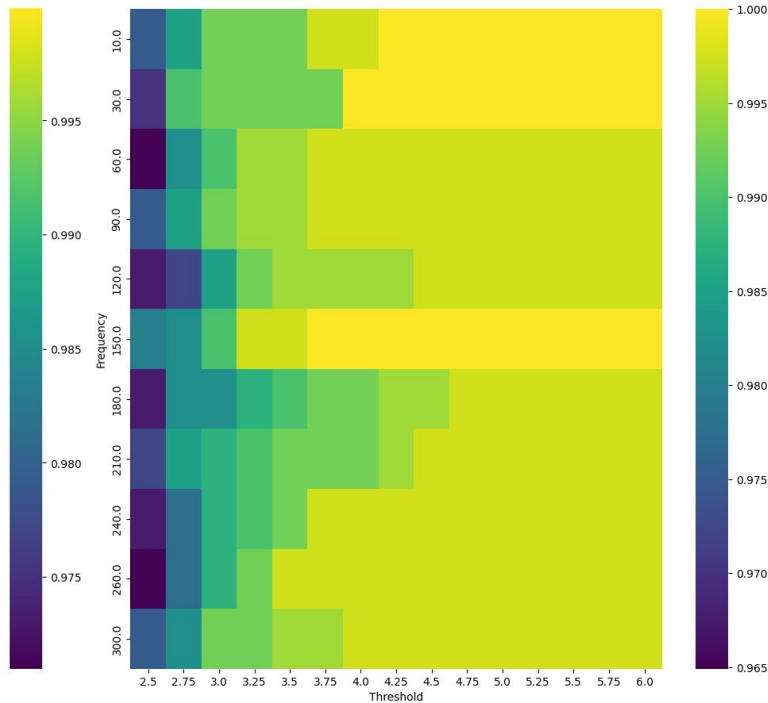
Look into heuristic score to find the most efficient exclusion strategy!

2D Optimization: SPE + MPE rate

- Metric used: SPE_MPE_Rate_difference_sigma
- Purpose: To catch statistically significant rate deviations



Livetime percentage



Background percentage

Background here is the UDTs in the B8 ROI passing some cuts like ETV, HSX and FVr.

Total background-tagged events across all days:
4411

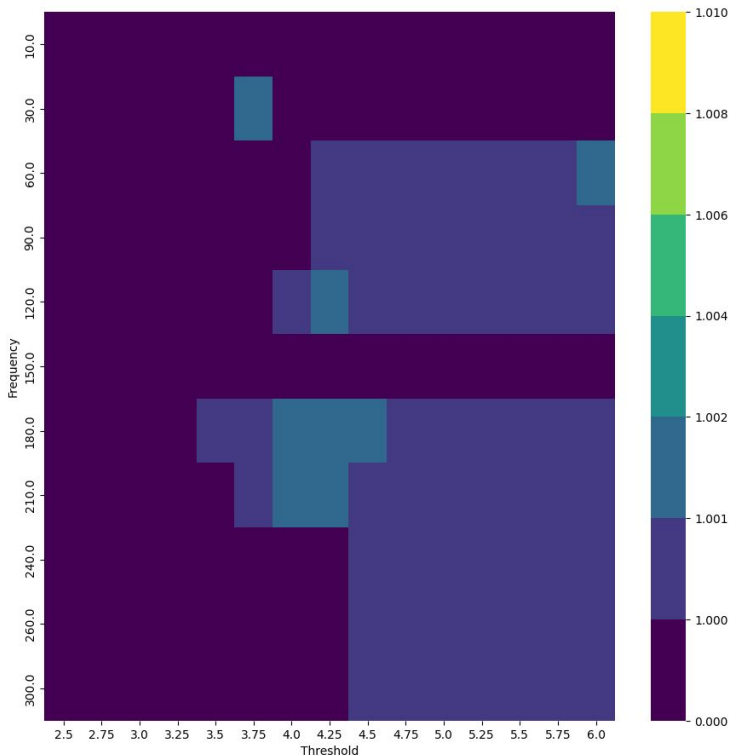
2D Optimization: SPE + MPE rate

Balancing Data Retention and Noise Rejection

$$\text{Heuristic} = \frac{\% \text{Livetime Remaining}}{\sqrt{\% \text{Background Remaining}}}$$

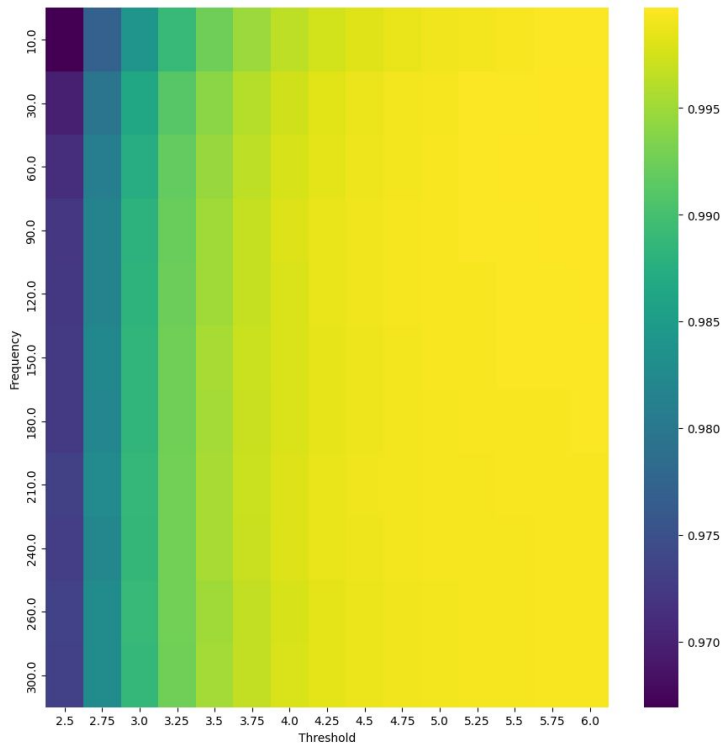
- Threshold: 4.25sigma
- Bin: 180s

- At 4.25σ and 180s binning a heuristic score (>1.002) is achieved
- Balances minimal livetime loss with effective background suppression

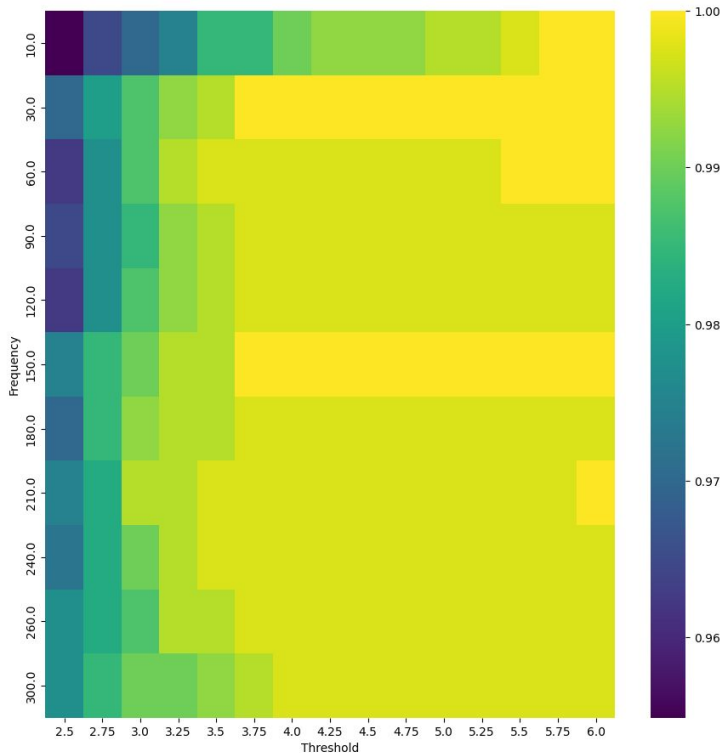


2D Optimization: S1 rate

Livetime percentage



Background percentage



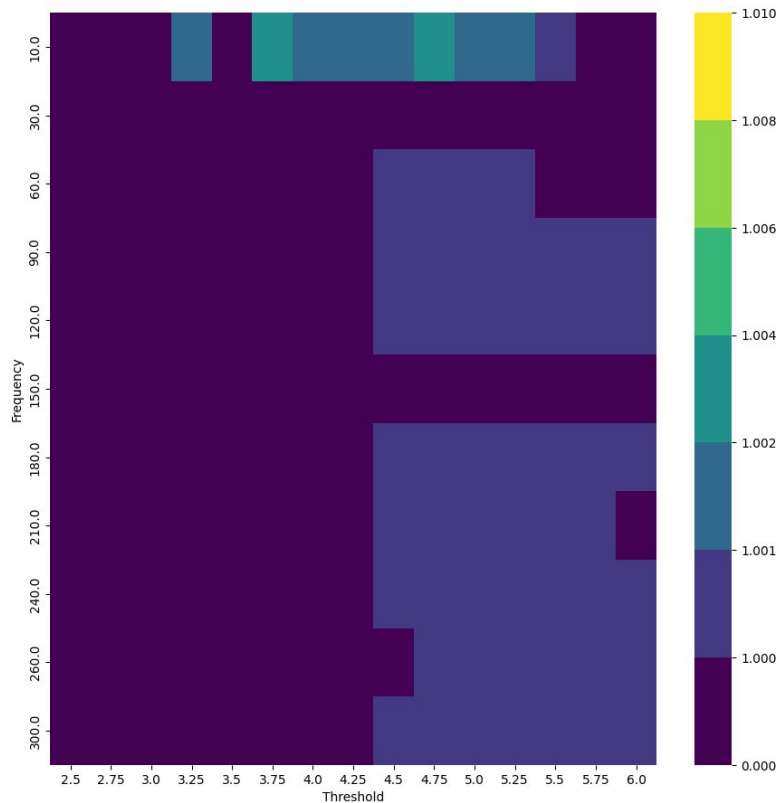
- Metric used: S1_Rate_difference_sigma
- Purpose: To catch statistically significant rate deviations

Background here is the UDTs in the B8 ROI passing some cuts like ETV, HSX and FVr.

2D Optimization: S1 rate

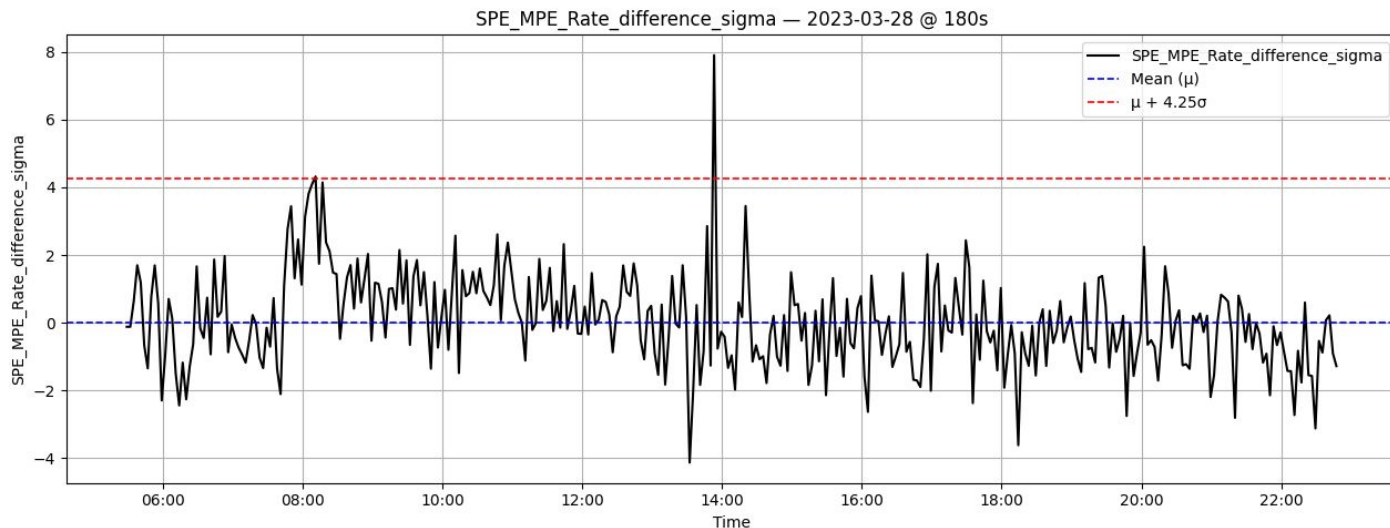
Balancing Data Retention and Noise Rejection

$$Heuristic = \frac{\%Lifetime\ Remaining}{\sqrt{\%Background\ Remaining}}$$



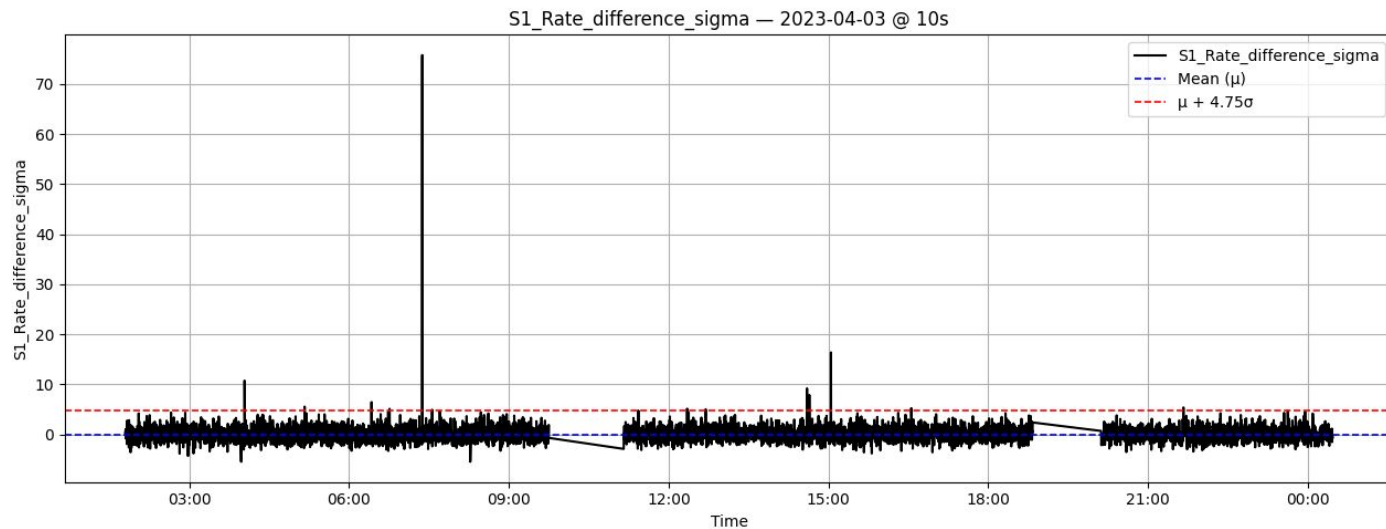
- Threshold: 4.75sigma
- Bin: 10s
- At 4.75σ and 10s binning a heuristic score (>1.002) is achieved
- Balances minimal lifetime loss with effective background suppression

Exclusion: SPE_MPE rate



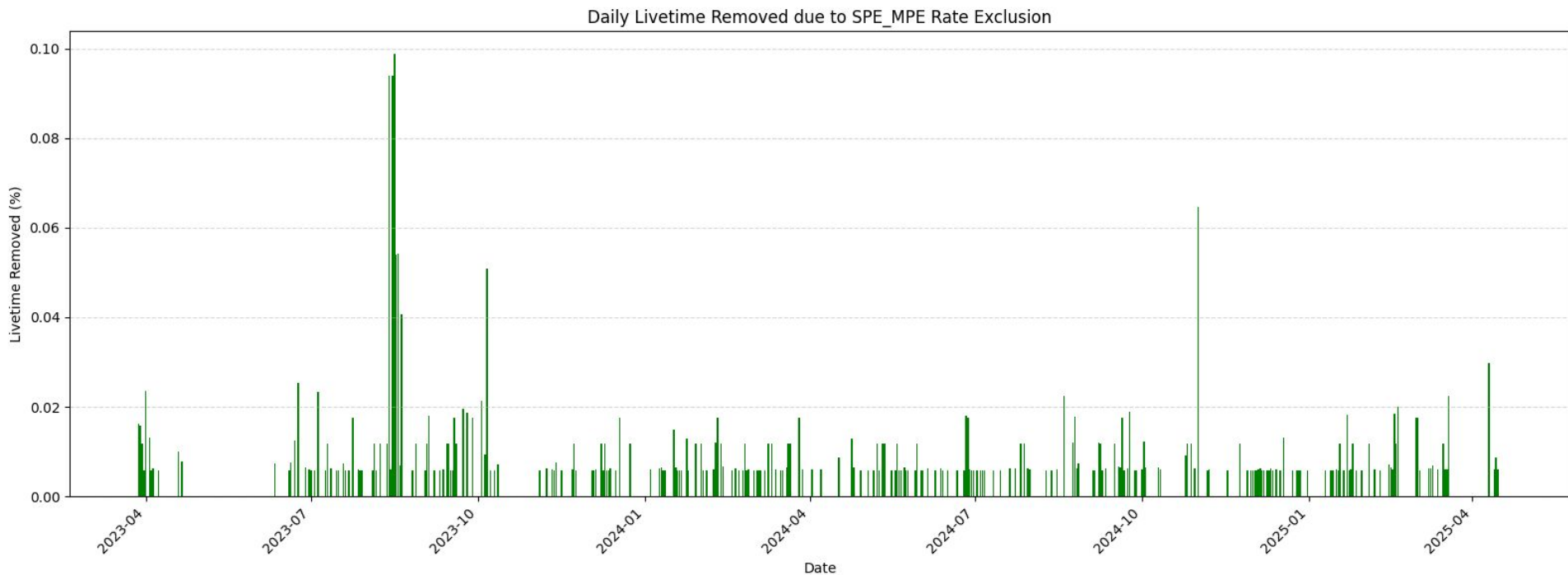
- 180s binning
- Exclude above 4.25sigma

Exclusion: S1 rate



- 10s binning
- Exclude above 4.75sigma

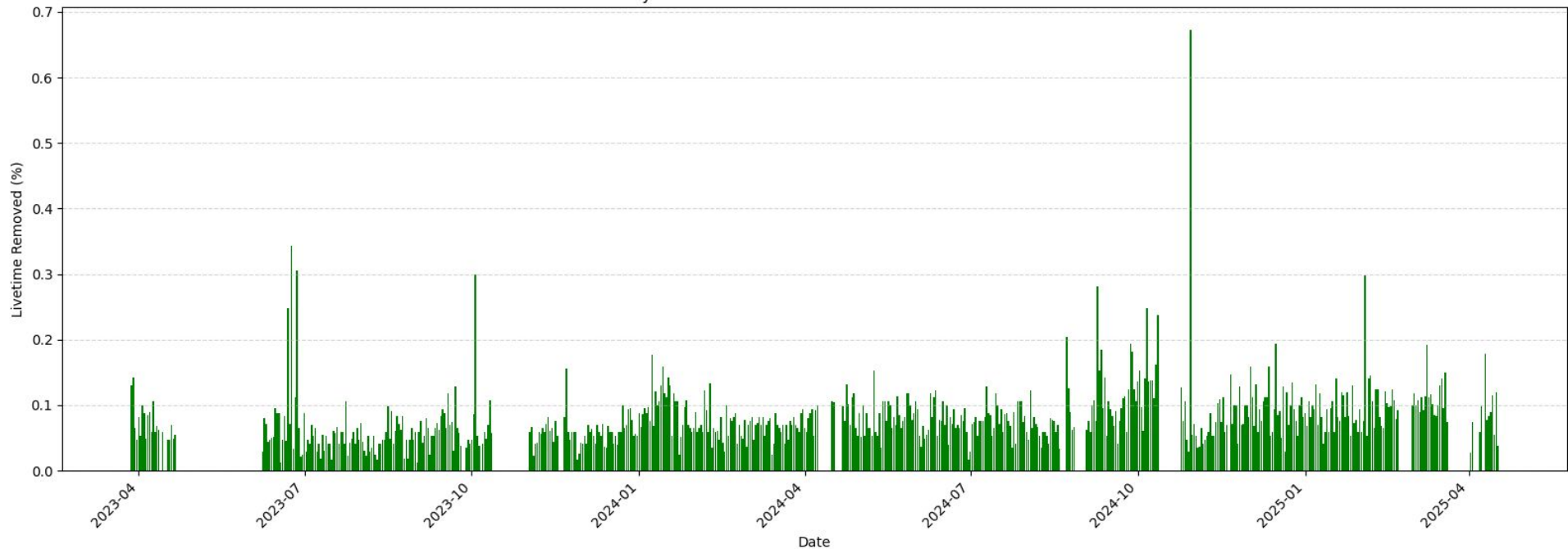
Livetime removed due to exclusion period: SPE_MPE rate



Bin width: 180s

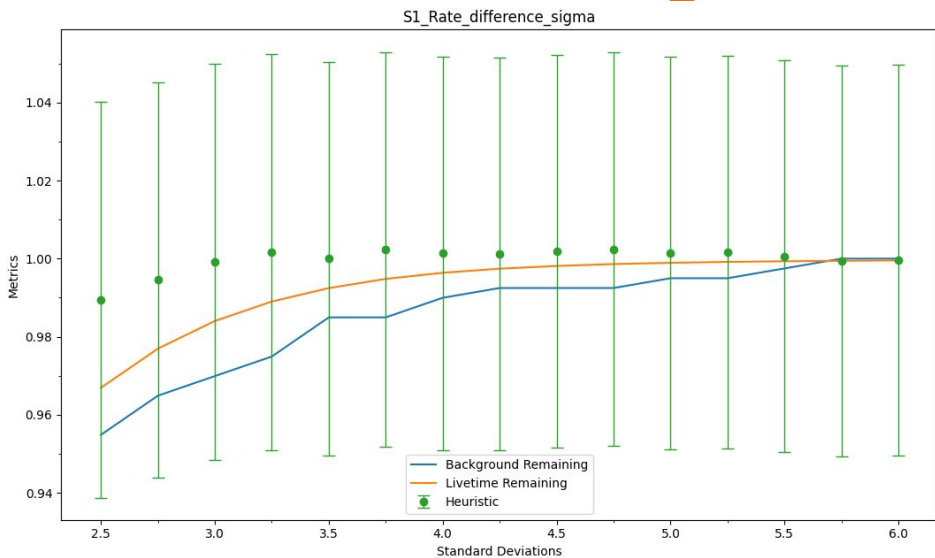
Livetime removed due to exclusion period: S1 rate

Daily Livetime Removed due to S1 Rate Exclusion



Bin width: 10s

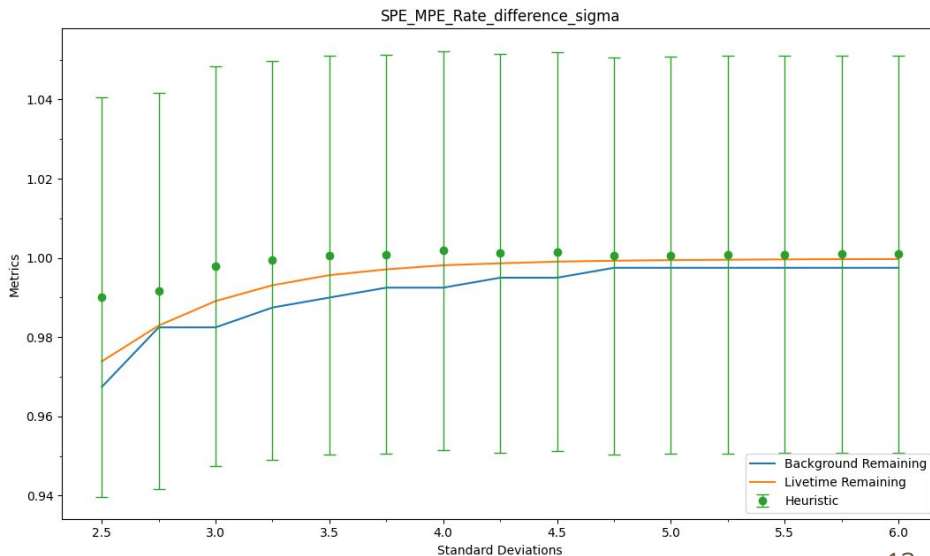
Effect of different sigma cutoffs:



- Threshold: 4.75sigma

- Bin: 10s

99.75% physics livetime remaining



- Threshold: 4.25sigma

- Bin: 180s

99.43% physics livetime remaining

Conclusion:

Optimized Exclusion:

- SPE + MPE rate cut:
 - 4.25σ threshold
 - Livetime retained: 99.43%
- S1 rate cut:
 - 4.75σ threshold
 - Livetime retained: 99.75%

Work contained in ALPACA merge request [!606](#)

Future Direction:

- Tuning event-level cuts

PMT Trip Exclusion:

Gavin C

PMT Trips

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- Cut spreadsheet: [link](#)
- Valid until: WS data through April 30, 2025
- ~46 hours calendar time excluded

Work contained in ALPACA merge request [!599](#)

Occasional rate spikes in the detector can cause PMTs to trip!

Cut Logic:

- Exclude intervals during which PMTs were tripped or powered down

Implementation:

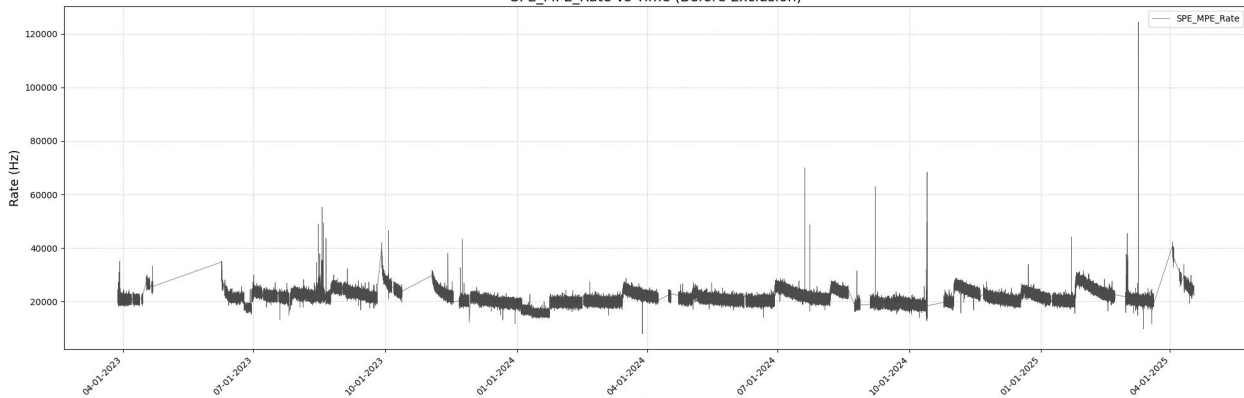
- Trip times were identified from ignition logs
- A 30-minute buffer was applied **before** each timestamp to conservatively bracket the affected periods

[Link to Gavin's slide](#)

Backup

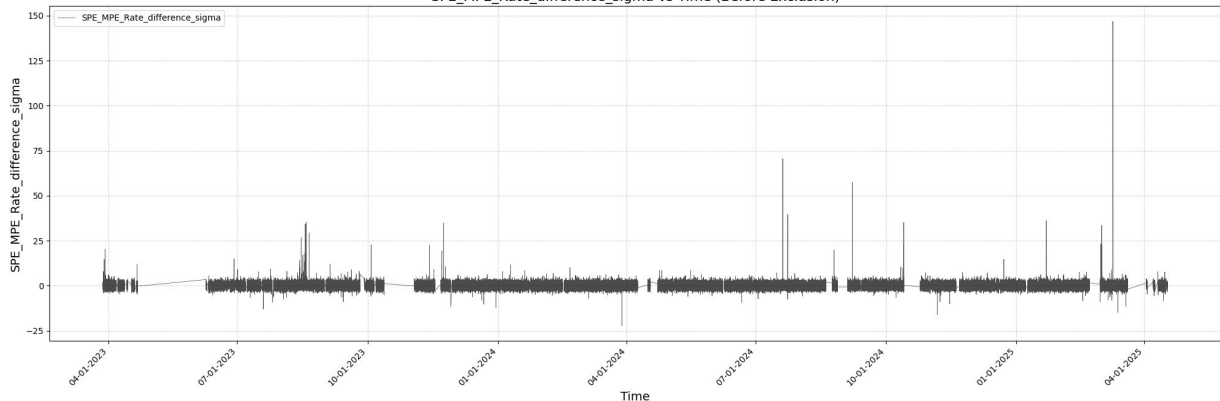
SPE+MPE rate & SPE+MPE rate difference sigma before exclusion:

SPE_MPE_Rate vs Time (Before Exclusion)



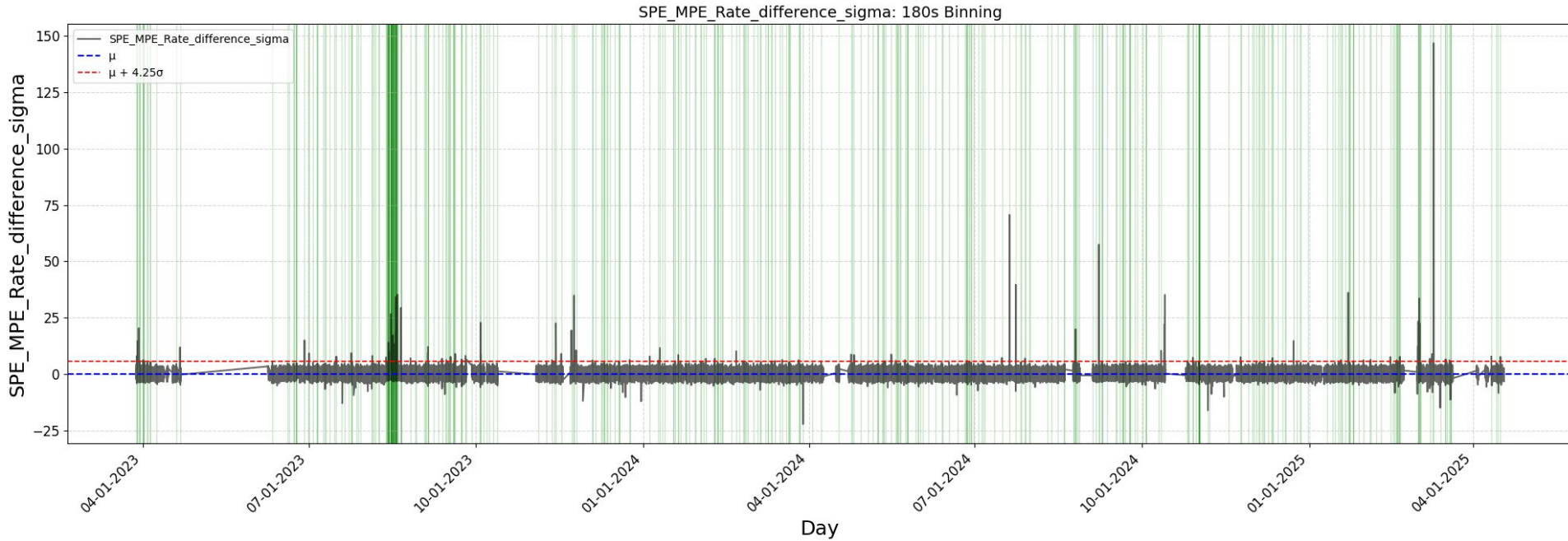
180s binning

SPE_MPE_Rate_difference_sigma vs Time (Before Exclusion)



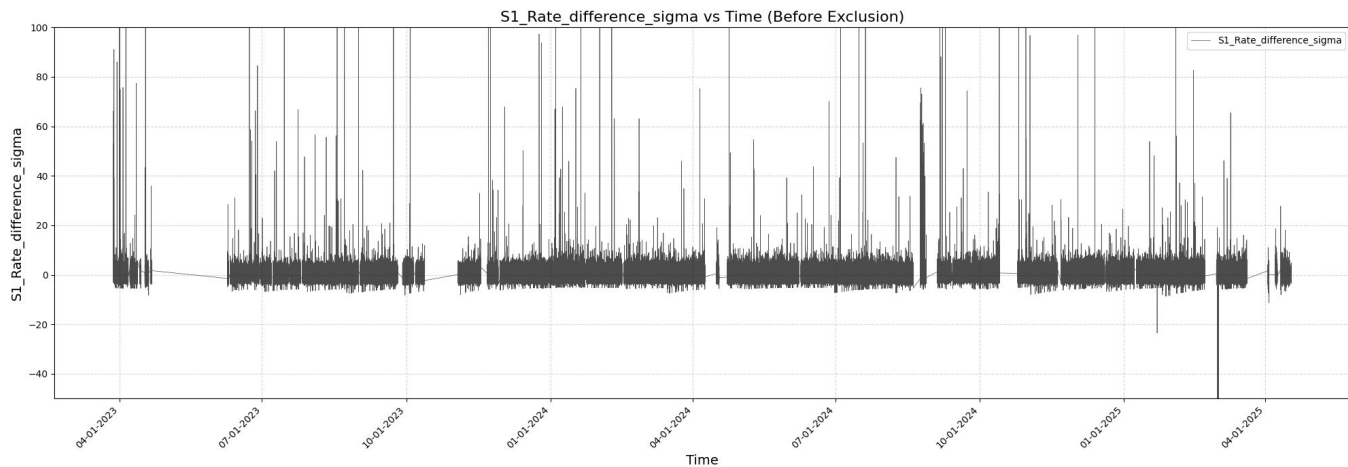
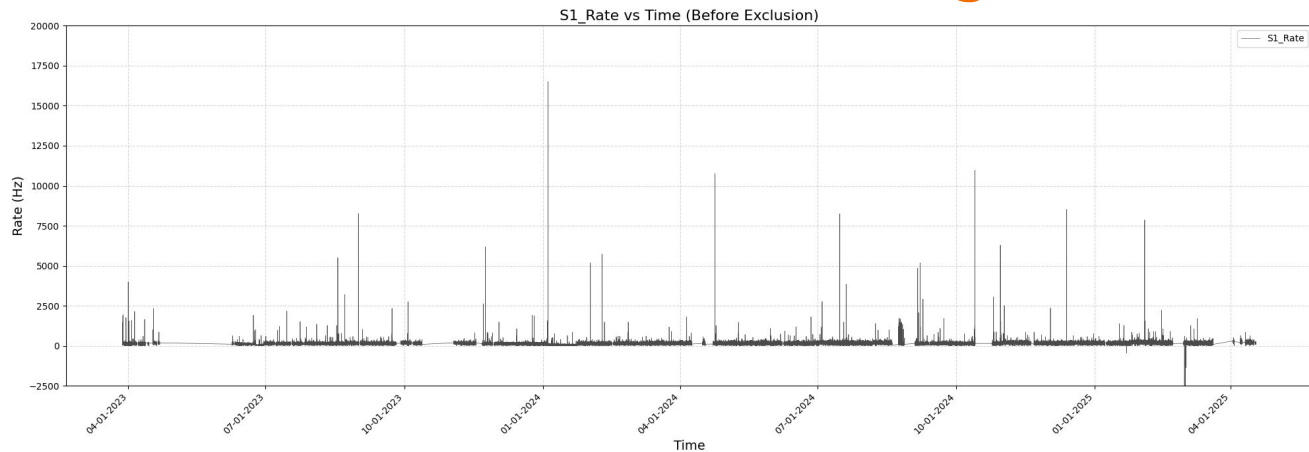
Overlaid exclusion period: SPE_MPE Rate Sigma Deviation (10s Binning)

Bin width: 180s



- Metric used: SPE_MPE_Rate_difference_sigma
- Purpose: To catch statistically significant rate deviations

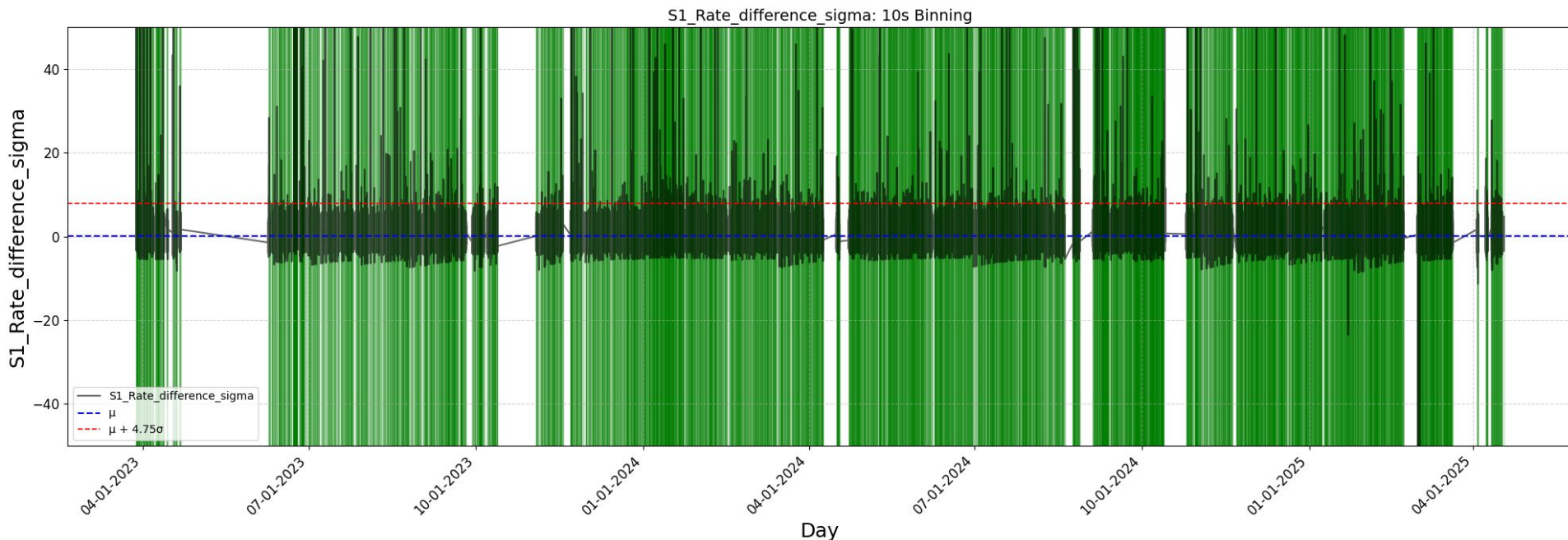
S1 rate & S1 rate difference sigma before exclusion:



10s binning

Overlaid Exclusion period: S1 Rate Sigma Deviation (10s Binning)

Bin width: 10s



Note: Due to the fine 10s binning and high density of data, exclusion periods (green regions) are difficult to visualize in this full-range plot. Slide 10 shows a zoomed-in one-day example where 4.75sigma threshold line indicates the region is getting excluded